Algorithms HW

1. A *circular polygon* $\mathcal{C}$ is a simply connected open set of $\mathbb{C}$ whose boundary is a union of circular arcs and is homeomorphic to a circle. Show that any Riemann map $\Delta \to \mathcal{C}$ extends to a homeomorphism $\overline{\Delta} \to \overline{\mathcal{C}}$.

Suppose for contradiction that a Riemann map $\Delta \to \mathcal{C}$ could not be extended continuously to the boundary. That is, we would have two sequences in Δ converging to some point on the boundary, $z_n, z_n' \to z_0 \in S^1$, such that their images converged to distinct points on $\partial\mathcal{C}$. In other words, there is some positive δ such that $|f(z_n) - f(z_m')| > \delta$ for all n, m.

Consider the following series of calculation. Consider circles α_r of radius r about z_0. For sufficiently large n, m and sufficiently small r there are points in both sequences which lie on α_r. Now, by the fundamental theorem of calculus we have

$$\begin{aligned}
\delta < \left|f(z_n) - f(z_m')\right| &= \left|\int_{\alpha_r} f'(\xi)d\xi\right| \\
&= \left|\int_{\theta_1(r)}^{\theta_2(r)} f'(z_0 + re^{i\theta})rd\theta\right| \\
&\leq \left|\int_{\theta_1(r)}^{\theta_2(r)}\right| \left|f'(z_0 + re^{i\theta})\right| rd\theta
\end{aligned}$$

where the second equality is given by parameterizing α_r and the final inequality given by the triangle inequality; the angles $\theta_1(r), \theta_2(r)$ are the angles on which z_n, z_m' respectively lie on α_r. Now, squaring both sides and then applying Cauchy-Schwarz (taking $h = fr^{1/2}$, $g = r^{1/2}$) gives

$$\begin{aligned}
\delta^2 &\leq \int \left|f'(z)\right|^2 rd\theta \int_{\theta_1(r)}^{\theta_2(r)} rd\theta \\
&\leq \int \left|f'(z)\right|^2 rd\theta \int_0^{2\pi} rd\theta \\
&\leq \int \left|f'(z)\right|^2 rd\theta \left(2\pi r\right).
\end{aligned}$$

Rearranging this inequality and then integrating over r gives

$$\int_{r=0}^{R} \frac{\delta^2}{r} dr \leq \int_r \int_\theta \left|f'(z)\right|^2 rd\theta dr = \int\int \left|f'\right|^2 dA = \text{Area}(f(\Delta)),$$

where the last equality is given by the fact that the Jacobian for the appropriate transformation is $|f'(z)|^2$. Now finally, notice that the left hand side diverges, but the right hand side is finite, since it is the area of a bounded open set. A contradiction, and so it must have been that f extended continuously to the boundary.

understand this jacobian statement, make a comment about the boundary being nice.

$\blacksquare$

2. Let $f : \Delta \to \Delta$ be a holomorphic function that extends continuously to $\overline{\Delta} \to \overline{\Delta}$, with $f(S^1) \subset S^1$. Show that f can be extended to a holomorphic function on $\mathbb{C}$ with finitely many points removed.

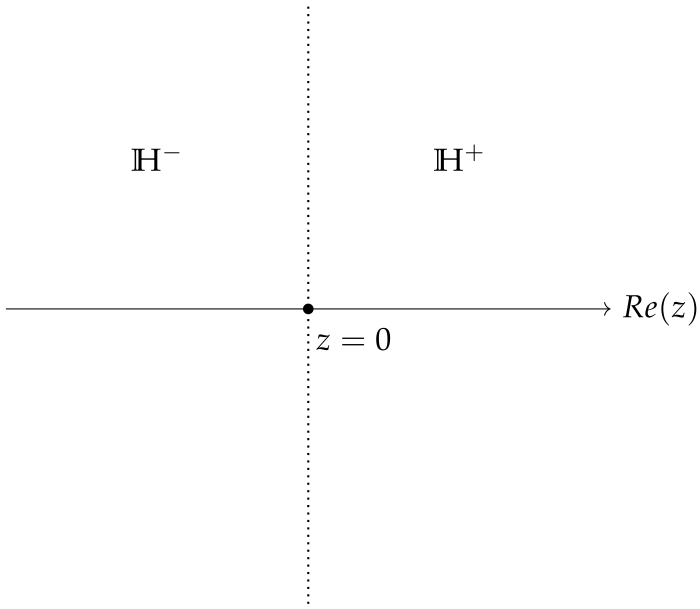
Recall that we have a biholomorphism from the $\Delta \to \mathbb{H}$ given by

$$h(z) = i \cdot \frac{1+z}{1-z}.$$

This map extends continuously to the boudaries $\bar{\Delta} \to \overline{\mathbb{H}} \cup \infty$ with $h(1) = \infty$. And so we can define a holomorphic function $g : \mathbb{H} \to \mathbb{H}$ by $g = h \circ f \circ h^{-1}$ which extends continuously to the actual boundary $\bar{\mathbb{H}} \cup \infty$ and $g(\bar{\mathbb{H}} \cup \infty) \subseteq \bar{\mathbb{H}} \cup \infty$. The preimages of 1 under f are those which get sent to infinity under g.

Suppose for a moment that we have a single point on the real line which gets sent to infinity. Say that we translate our coordinates so that this point is $z = 0$. Let us denote the "right half upper plane" by $\mathbb{H}^+ := \{z \in \mathbb{H} : Re(z) > 0\}$ and similarly let $\mathbb{H}^-$ denote the "left upper half plane".

Now, the restriction of g to $\mathbb{H}^+$ defines a holomorphic function which extends continuously to the interval $(0, \infty)$ with the image of $(0, \infty) \subseteq \mathbb{R}$. The Schwarz reflection principle then allows us to extend $g|_{\mathbb{H}^+}$ to a holomorphic function on the entire right half plane $\mathbb{R}^+ := \{z \in \mathbb{C} : Re(z) > 0\}$. The same argument means that we can extend $g|_{\mathbb{H}^-}$ to a holomorphic function on the entire left half plane. We now have a holomorphic function $\mathbb{C} \setminus L \to \mathbb{C}$ where L is the vertical line through $z = 0$. Let us denote this function by $\tilde{g}$.



We can extend $\tilde{g}$ further to a function $G : \mathbb{C} \setminus \{z = 0\} \to \mathbb{C}$ in the following way

$$G(z) = \begin{cases} \tilde{g}(z) & z \notin L \\ g(z) & z \in L \cap \mathbb{H} \\ \overline{g(\bar{z})} & z \in L \cap -\mathbb{H}. \end{cases}$$

The proof that Scharz reflection gives a holomorphic function also proves that this function is holomorphic as a function $\mathbb{C} \setminus \{z = 0\} \to \mathbb{C}$.

We can translate G into a holomorphic extension of f by conjugating $h^{-1} \circ G \circ h : \mathbb{C} \setminus \{p\} \to \mathbb{C}$ where p is the preimage of 1 under the continuous extension $\overline{f}$.

If f has more than one point on S^1 which is sent to 1 (i.e. more than one point on $\mathbb{R}$ is sent to ∞ under h) then we can repeat a similar procedure, except now we Schwarz reflect vertical strips bounded by the vertical lines through the points on the real line which are sent to ∞ by h.

Lastly, must there only be finitely many points?

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4. Find two bounded open subsets of $\mathbb{C}$ that are homeomorphic but not biholomorphic.

We know from the Riemann mapping theorem that any simply connected open set (which is not all of $\mathbb{C}$) is biholomorphic to the unit disc. And so, any two bounded simply connected open sets are biholomorphic to each other. Thus, we need at least one of our open sets to be non-simply connected. Moreover, since we want to find a homeoomrphism between our open sets, it follows then that both are not simply connected (and much have the same homology). (From here, how do we show that there's no biholomorphism between two given open sets) ∎

4

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